

Alexander David Suer

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Education

- Doctor of Philosophy, Mechanical Engineering, University of Cincinnati, Cincinnati, OH** **Expected: August 2027**
- GPA: 3.91/4.00
- Master of Science, Mechanical Engineering, University of Cincinnati, Cincinnati, OH** **May 2025**
- GPA: 3.91/4.00
- Bachelor of Science, Mechanical Engineering, University of Cincinnati, Cincinnati, OH** **April 2022**
- GPA: 3.80/4.00
 - Robotics and Automation Minor

Research

- Center for Intelligent Metrology & Sensing, University of Cincinnati, Cincinnati, OH** **May 2022 – Present**
- Developed robotic perception software for automated disassembly using 3D computer vision, point cloud processing, RGBD cameras, structured-light scanners, and sensor fusion to enable robot-arm fastener detection and removal.
 - Pretrained a PINN Neural Operator model in PyTorch for prediction of medicine diffusion in the human eye. Prediction time reduced by 10 times from traditional FDM model with similar accuracy.
 - Created a UNet CNN for artery/vein semantic segmentation of eye fundus images for health risk assessment.
 - Optimized ensemble model for rock drill fault detection and classification with accuracy of 99.3% for the Prognostics and Health Management Society's 2022 European data challenge.
 - Created a signal processing framework using XGBoost, achieving F1 score of 0.90, for selecting best features for the Prognostics and Health Management Society's 2022 North American data challenge.

Publications

- [J1] **A. Suer**, Q. Pan, T. Yahia, G. Marasini, K. Onaga, K. Haubeil, O. Adekoya, J. Dong, X. Jia. "Novel Methodology for 3D Vision-Based Screw Pose Estimation and Robot-Assisted Screw Removal," *Measurement*. (submitted 2026)
- [C1] **A. Suer**, X. Han, X. Jia, J. Moyne, F. Li, A. Shankarraman. "Improving Fault Detection Across Chambers by Adding Feature Informed Dynamic Time Warping to Transfer Learning," *APC-SM Conf.*, Oct. 2025.
- [C2] **A. Suer**, X. Han, M. Miller, X. Jia, J. Moyne, J. Iskandar, F. Li, A. Shankarraman, "Accuracy Improvements for Transfer Learning Across Recipes Applied to Sensor Trace Data," *APC-SM Conf.*, Oct. 2024.
- [C3] **A. Suer**, X. Yang, R. Yang, X. You, Z. Zhang, and J. Dong, "Development of an Autonomous Robotic Snowplow for Residential Use," presented at the ASME 2023 International Mechanical Engineering Congress and Exposition, American Society of Mechanical Engineers Digital Collection, Feb. 2024. doi: 10.1115/IMECE2023-114010.
- [C4] X. Han, **A. Suer**, M. Miller, X. Jia, J. Moyne, F. Li, J. Iskander, A. Shankarraman. "A Novel Trace Transfer Learning Framework to Improve Dataset Quality and Reduce Modeling Effort," *2026 37th Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC)*, 2026.
- [J2] X. Han, **A. Suer**, B. Foreman, and X. Jia, "Topological Visualization of Intracranial Pressure Morphology Variations and Real Time Data Trajectory Mapping," *IEEE J. Biomed. Health Inform.*, pp. 1–8, 2025, doi: 10.1109/JBHI.2025.3621411.
- [J3] Y. Su, J. Shi, Y.-M. Hsu, D.-Y. Ji, **A. D. Suer**, and J. Lee, "Volumetric nondestructive metrology for 3D semiconductor packaging: A review," *Measurement*, vol. 225, p. 114065, Feb. 2024, doi: 10.1016/j.measurement.2023.114065.
- [C5] K. Haubeil, T. Yahia, **A. Suer**, D. Siegel, D. Davis, and X. Jia, "A Novel Methodology for Vision Backbone Network Fine-Tuning and Continual Learning in Optical Inspection Tasks," *Annual Conf. of the PHM Society*. 2025.
- [C6] T. Yahia, K. Haubeil, **A. Suer**, Y. Qu, J. Dong, and X. Jia, "A Novel 3D Sensing Framework for Safety Monitoring in Human-Robot Collaboration Work Cells," *Annual Conf. of the PHM Society*. 2025.
- [C7] M. Miller, V.V. Kalaiarasan, **A. Suer**, X. Jia, J. Moyne, J. Iskandar, and F. Li. "Applying Transfer Learning Across Chambers to Improve Model Quality and Reduce Modeling Effort," *APC-SM Conf.*, Oct. 2023.
- [C8] T. Minami, **A. Suer**, P. Kundu, S. Siahpour, and J. Lee, "Novel Ensemble Domain Adaptation Methodology for Enhanced Multi-class Fault Diagnosis of Highly-Connected Fleet of Assets," *PHM Soc. Asia-Pac. Conf.*, vol. 4, no. 1, Art. no. 1, Sept. 2023, doi: 10.36001/phmap.2023.v4i1.3701.
- [C9] J. Taco, P. Gore, T. Minami, P. Kundu, **A. Suer**, and J. Lee, "A Novel Methodology for Health Assessment in Printed Circuit Boards," *PHM Soc. Eur. Conf.*, vol. 7, no. 1, Art. no. 1, June 2022, doi: 10.36001/phme.2022.v7i1.3308.

Experience

Graduate Researcher, *University of Cincinnati with Applied Materials* **September 2022 - Present**

- Designed dynamic time warping based trace data transfer learning method to strengthen datasets for fault detection. Improved T2-SPE fault detection AUC to 0.973 from 0.625 on Eigenvector Metal Etch Dataset.

Researcher, *P&G Digital Accelerator, Cincinnati, OH* **January - May 2019, September 2024 – May 2025**

- Designed point cloud processing pipeline and an accompanying visualization tool for verifying packaging geometry with average accuracy over 88%.
- Created an ASP.NET web application with Qualtrics' APIs that generates P&G consumer product review surveys.
- Designed and programmed an Android application with an SQLite database for perfume chemists to create and edit perfume recipes.

Graduate Researcher, *University of Cincinnati with Mitsubishi Electric* **September 2022 – March 2023**

- Designed and benchmarked a transfer learning model for robot fault detection with over 90% online accuracy.

Software Development Co-op, *International TechneGroup (Wipro), Milford, OH* **January – August 2021**

- Designed new diagnostics that check CAD models for geometric and annotation defects in the CADIQ software.
- Designed and implemented improvements to CADIQ's Qt based UI menus.

R&D Co-op, *SETCO Sales Company, Cincinnati, OH* **September - December 2018, May – August 2019**

- Created 3D models of new spindle parts in Inventor. Tested designs under theoretical operating conditions with FEA. Accepted or rejected designs for prototyping based on whether they achieved design requirements.
- Constructed and conducted endurance tests on new products by measuring coolant flowrate and product temperature data. Rejected product designs if temperature and flowrate were outside criteria.
- Drafted part and assembly AutoCAD drawings of new products using ISO standards. Drawings of parts were sent to manufacturers for production.

Activities

Treasurer, *UC Robotics Team, UC, Cincinnati, OH* **October 2017 – December 2022**

- Designed and programmed a 3D model of the team's IGVC competition robot in SolidWorks and simulated it in Gazebo to fine tune of the robot's navigation stack.
- Developed a ROS1 node for object detection and visual odometry with a fisheye camera.
- Designed and assembled capstone project robot snowplow's electronics panels.

Member, *Engineers without Borders, UC, Cincinnati, OH* **March 2020 – May 2025**

- Planned additions to the sanitary system of the Tanzanian village Nyambogo. Estimated pipeline flow and made drawings of pipe network around the system's water tanks.

Skills

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- **Programming:** Proficient in Python, PyTorch, Docker, MATLAB, Java, C#, ROS, Streamlit, Qt, and Git.
 - **Software:** Proficient in Solidworks, NX, AutoCAD, and Inventor.

Awards

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- Rindsberg Fellowship, National Merit, and Cincinnati Excellence Scholarships Recipient
 - Best Student Paper: Advanced Process Control-Smart Manufacturing Conference 2023